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ORIGINAL ARTICLE

Occurrence of the genes encoding carbapenemases, ESBLs and class 1 integron-integrase among fermenting and non-fermenting bacteria from retail goat meat

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Significance and impact of the study: The emergence of carbapenemases and extended-spectrum β-lactamases (ESBLs) in bacteria has now become a global problem leading to failure of advanced antimicrobial therapy. We report for the first time the *bla*$_{\text{NDM-1}}$ gene in non-fermenting Gram-negative bacterial isolates, and the *bla*$_{\text{KPC-2}}$ gene in an *Escherichia coli* isolate from goat meat in India. The presence of carbapenemase genes in bacteria from goat meat develops serious concerns about public health and food safety. Analysis of genetic determinants of β-lactam resistance in bacteria would be helpful to formulate a suitable control strategy.

Keywords
antimicrobial resistance, carbapenemase, ESBL, goat meat, Gram-negative bacteria, integron, public health.

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Abstract

The present study was planned to detect the genes encoding carbapenemases, ESBLs and class 1 integron-integrase among bacteria obtained from retail goat meat. Fermenting and non-fermenting bacterial isolates (n = 57), recovered from 61 goat meat samples, were identified by 16S rRNA gene sequencing. Antimicrobial susceptibility of isolates was tested by the broth dilution method using ceftazidime, cefotaxime, meropenem and imipenem. Plasmids were isolated and tested for their physical characters. Plasmids were subjected to screening of carbapenemase, ESBL and *intI1* gene. Conjugation assay was performed using *bla*$_{\text{NDM}}$-positive isolates as the donor, and *Escherichia coli* HB101 as the recipient. Isolates showed the high rates of resistance to ceftazidime (77·2%), cefotaxime (70·2%), meropenem (22·8%) and imipenem (17·5%). They showed variability in number and size (~1 to >20 kb) of plasmids. Among all, 1, 4, 13 and 31 isolates showed the *bla*$_{\text{KPC}}$, *bla*$_{\text{NDM}}$, *bla*$_{\text{SHV}}$ and *bla*$_{\text{TEM}}$ genes, respectively. The *bla*$_{\text{KPC-2}}$ gene was observed in one *E. coli* isolate. The *bla*$_{\text{NDM-1}}$ gene was detected in *Stenotrophomonas maltophilia* (n = 2), *Acinetobacter baumannii* (n = 1) and *Ochrobactrum anthropi* (n = 1) isolates. These isolates co-harboured the *bla*$_{\text{TEM}}$ and *bla*$_{\text{SHV}}$ genes. The *intI1* gene was detected in 22 (38·6%) isolates, and 16 of these isolates showed the carbapenemase and/or ESBL genes. The conjugative movement of *bla*$_{\text{NDM}}$ gene could not be proved after three repetitive mating experiments. The presence of genes encoding carbapenemases and ESBLs in bacteria from goat meat poses public health risks.

Introduction

The emergence of antimicrobial resistance in Gram-negative bacteria is one of the foremost problems that needs utmost attention. The β-lactam antibiotics have been consistently used worldwide since last six decades for the treatment and control of bacterial infections in human and veterinary medicine. However, their effectiveness is severely limited due to the emergence of β-lactam-resistant bacteria in humans, animals and the environment (Allen *et al.* 2010; Woodford *et al.* 2014; Singh *et al.* 2017). The most common transferable antimicrobial resistance is developed by means of bacterial β-lactamases such as extended-spectrum β-lactamases (ESBLs), AmpC β-lactamases and carbapenemases which mediate the deactivation of β-lactam antibiotics (Lupo *et al.* 2012). Carbapenems have been considered the therapy of 'last resort' to manage the infections caused by multidrug-resistant (MDR) bacteria (Morrill *et al.* 2015). Unfortunately, during the past decades, MDR- and carbapenem-resistant *Enterobacteriaceae* isolates have been emerged widely, restricting this effective therapy (Pournaras *et al.* 2016). New Delhi metallo-β-lactamase (NDM) enzymes, associated with multidrug resistance in bacteria, first reported in 2008, are now found worldwide (Kumarasamy *et al.* 2010; van Duin and Doi 2017). Besides *Enterobacteriaceae*, the acquired carbapenemase genes have been observed in non-fermenting Gram-negative bacteria such as *Stenotrophomonas maltophilia*, *Acinetobacter* spp. and *Ochrobactrum anthropi* (Scotta *et al.* 2011; Zhang *et al.* 2013a; Jones *et al.* 2014). A number of previous studies have reported the occurrence of carbapenemases among *Enterobacteriaceae* isolates in food animals (including livestock and poultry) or in food animal products from various countries such as Algeria, China, Egypt, Germany, India, Italy, Lebanon, Romania and USA (Madec *et al.* 2017; Bonardi and Pitino 2019). Increasing reports on carbapenemase-producing *Enterobacteriaceae* (CPE)/carbapenemase-producing non-*Enterobacteriaceae* (CPnE) isolates attract attention to carry out antimicrobial resistance analysis of food animal products, since these animal products are directly linked with the health of humans and animals. In India, CPE and/or CPnE have been reported from the environmental sources and human patients (Kumarasamy *et al.* 2010; Jones *et al.* 2014). It is not known, however, whether or not the food animal products are contaminated with these CPE and CPnE. Therefore, the present study was intended to detect and characterize the carbapenemase and ESBL genes among bacteria in retail goat meat from Chhattisgarh state in India.

Results and discussion

Morphologically dissimilar and discrete bacterial colonies from HiCrome ESBL agar medium were kept in pure culture slants and recognized as distinct isolates. These isolates were specifically identified based on their 16S rRNA gene sequences and grouped as *Escherichia coli* (n = 29), *Klebsiella pneumoniae* (n = 13), *Citrobacter freundii* (n = 07), *Acinetobacter baumannii* (n = 03), *S. maltophilia* (n = 02), *Klebsiella oxytoca* (n = 01), *Citrobacter koseri* (n = 01) and *O. anthropi* (n = 1). They included *Enterobacteriaceae* (fermenting, n = 51) and non-*Enterobacteriaceae* (non-fermenting, n = 6) bacterial isolates. Thus, a total of 57 Gram-negative isolates were identified from 61 goat meat samples. ESBL-producing *Enterobacteriaceae* bacteria have been isolated using HiCrome ESBL agar in previous studies (Rameshkumar *et al.* 2015; Tepeli and Demirel Zorba 2017).

Coliform (*Enterobacteriaceae*) contamination of meat could be due to soiling of carcass surface by intestinal contents (Tanganyika *et al.* 2017; Diyantoro and Wardhana 2019). Non-fermenting Gram-negative bacteria such as *S. maltophilia*, *A. baumannii* and *O. anthropi* might be acquired from nonhuman environmental sources (Berg *et al.* 2005; Martínez 2008). They have been evolved as emerging opportunistic pathogens associated with nosocomial infections, especially in immunocompromised patients (Higgins *et al.* 2001a; Sanchez 2015).

Antimicrobial susceptibility testing (AST) of isolates revealed high resistance rates to third-generation cephalosporins such as ceftazidime (77·2%) and cefotaxime (70·2%) with a minimum inhibitory concentration (MIC) of >16 μg ml^{-1}. Resistance to carbapenems such as meropenem (22·8%) and imipenem (17·5%) was also recorded (Fig. 1). Among non-*Enterobacteriaceae* isolates, all six isolates were resistant to ceftazidime and cefotaxime, while four of them were also resistant to meropenem and imipenem (Table 1).

Beta-lactam resistance in Gram-negative bacteria usually occurs through the production of β-lactamases including ESBLs and carbapenemases (Lupo *et al.* 2012; Madec *et al.* 2017; Bonardi and Pitino 2019). The reduced susceptibility of *S. maltophilia* to antibiotics might be associated with the intrinsic resistance factors such as low membrane permeability, the presence of multidrug resistance efflux pumps and the antibiotic-

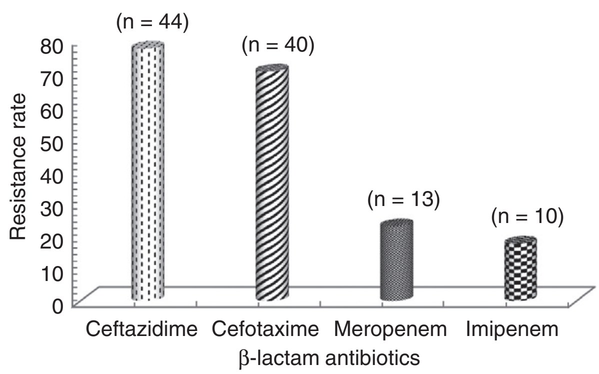

Figure 1 The rates of β-lactam resistance among bacterial isolates (n: number of isolates).

Table 1 Antimicrobial resistance of carbapenemase-producing non-*Enterobacteriaceae* isolates

	MIC (μg ml^{-1})			
Bacterial isolate	Ceftazidime	Cefotaxime	Meropenem	Imipenem
Acinetobacter baumannii (01)	>128	>128	>64	>32
Stenotrophomonas maltophilia (02)	>128	>128	>32	>16
Ochrobactrum anthropi (01)	>128	>256	>32	>32

modifying enzymes (Sanchez *et al.* 2009). Moreover, development of antimicrobial resistance in these bacteria might be attributed to mutations in addition to horizontal transfer of antimicrobial resistance genes (Sanchez 2015). Similarly, the *O. anthropi* and *A. baumannii* strains express a broad drug-resistant spectrum through the production of ESBLs and carbapenemases (Higgins *et al.* 2001b; Scotta *et al.* 2011; Zhang *et al.* 2013a).

There was variability in the number and size of plasmids among bacterial isolates. Most of the isolates (43) harboured one to four plasmids. Moreover, one *E. coli* isolate carried a cluster of plasmids (seven to eight) ranging from about 1 kb to less than 20 kb in size. However, 13 (22·8%) isolates did not show any visible plasmid. Majority of the isolates harboured 2- to 20-kb-sized plasmids. Larger plasmids of more than 20 kb were also recorded in seven isolates including five of *E. coli* and one of each of *C. freundii* and *A. baumannii*. There were variable-sized plasmids among isolates of the same genus. Moreover, the isolates of different genera carried plasmids of identical size. Diversity of plasmids among bacterial isolates indicates that these bacteria might have evolved and spread in different niches. In contrast, the presence of identical-sized plasmids among isolates might be due to their clonal spread. The plasmids often encode functions giving extra advantages to their hosts in presence of selective pressure. Among bacteria, plasmids vary greatly in their physical characters. A single bacterium can acquire several plasmids from multiple donors in different environmental conditions, and thus, poses serious risks to public health. The similar features of plasmids among bacteria from environmental sources have been reported earlier (Kumarasamy *et al.* 2010; Singh *et al.* 2019).

Collectively, a total of 32 (56·1%) bacterial isolates harboured the resistance genes encoding carbapenemases and/or ESBLs. The *bla*$_{TEM}$ was the most commonly detected *bla* gene among the isolates. It, alone or in combination with *bla*$_{SHV}$ or *bla*$_{NDM}$, was detected in 31 isolates (54·4%) (Fig. 2). Out of 57 isolates, 13 (22·8%) isolates showed the *bla*$_{SHV}$ gene. Among the *bla*$_{SHV}$-positive isolates, nine isolates co-harboured the *bla*$_{TEM}$ gene, while four isolates co-harboured the *bla*$_{TEM}$ and *bla*$_{NDM}$ gene. The co-production of *bla*$_{SHV}$ and *bla*$_{TEM}$ gene was

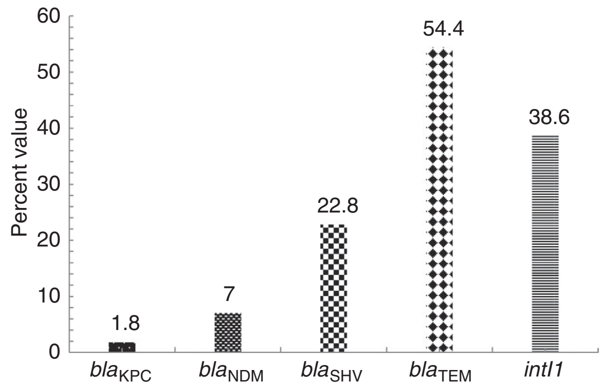

Figure 2 Occurrence of the genes encoding carbapenemases, ESBLs and class 1 integron-integrase among bacterial isolates.

recorded in eight isolates of *Klebsiella* spp. However, it was observed only in one isolate of *E. coli*.

The *bla*$_{NDM}$ gene was detected in four isolates including both of *S. maltophilia* and one of each of *A. baumannii* and *O. anthropi* (Table 2). These isolates also carried the *bla*$_{TEM}$ and *bla*$_{SHV}$ genes. *S. maltophilia* isolates did not show any visible plasmid band in agarose gel. The *bla*$_{NDM}$-harbouring *A. baumannii* isolates possessed four visible plasmids measuring about 2, 5, 20 kb and more than 20 kb in size, while the *bla*$_{NDM}$-harbouring *O. anthropi* isolate possessed a single plasmid of about 20 kb. None of the coliform isolates was found to harbour the *bla*$_{NDM}$ gene. However, one *E. coli* (O2 serotype) isolate harboured the *bla*$_{KPC}$ gene. This isolate was found to carry a single plasmid of about 20 kb.

Among carbapenem-resistant isolates (n = 16), the carbapenemase genes *bla*$_{NDM}$ and/or *bla*$_{KPC}$ were detected only in five isolates. In the absence of *bla*$_{NDM}$ or *bla*$_{KPC}$ gene, carbapenem resistance in bacterial isolates might be due to the other possible reasons. Production of cephalosporinase (AmpC β-lactamases and ESBLs have low-level carbapenemase activity) can also develop carbapenem resistance in enterobacteria (Clinical and Laboratory Standards Institute 2015). Moreover, the efflux pump may also play an important role in developing carbapenem resistance in bacteria (Pruthvishree *et al.* 2017). Furthermore, the presence of other carbapenemase genes could not be ruled out.

Table 2 Bacterial isolates and their antimicrobial resistance profile

Isolate code	Species of the isolate (n)	Resistance to antimicrobials	Plasmid (n)	*bla*/ *intI1* gene
303b	*Citrobacter freundii* (01)	CAZ, CTX	+ (1)	bla_{TEM}
307b, 314b/L	*Klebsiella pneumoniae* (02)	CAZ, CTX	+ (1)	bla_{TEM}, *intI1*
308a, 337a, 354a	*Escherichia coli* (UT)* (03)	CAZ, CTX	+ (2)	bla_{TEM}, *intI1*
308b, 315b, 349b	*Klebsiella pneumoniae* (03)	CAZ, CTX	NV	bla_{TEM}
309a, 342a, 360a	*Escherichia coli* (UT)* (03)	CAZ, CTX	+ (1)	bla_{TEM}
326a	*Escherichia coli* (UT)* (01)	CAZ, CTX	+ (4)	bla_{TEM}, *intI1*
339b	*Citrobacter freundii* (01)	CAZ, CTX, MRP, IPM	+ (4)	bla_{TEM}, *intI1*
342b	*Klebsiella oxytoca* (01)	CAZ, CTX	+ (2)	bla_{TEM}, bla_{SHV} *intI1*
343b, 344b, 345b, 346b, 348b	*Klebsiella pneumoniae* (05)	CAZ, CTX	+ (1)	bla_{TEM}, bla_{SHV}
347b	*Klebsiella pneumoniae* (01)	CAZ, CTX	+ (2)	bla_{TEM}, bla_{SHV}
348a	*Escherichia coli* (O2)* (01)	CAZ, CTX, MRP, IPM	+ (1)	bla_{KPC}, *intI1*
351a	*Escherichia coli* (O2)* (01)	CAZ, CTX	+ (3)	bla_{TEM}, *intI1*
352a, 355a, 356a	*Escherichia coli* (R/UT)* (03)	CAZ, CTX	NV	bla_{TEM}
355b	*Klebsiella pneumoniae* (01)	CAZ, CTX, MRP, IPM	+ (4)	bla_{TEM}, bla_{SHV}, *intI1*
357a	*Escherichia coli* (UT)* (01)	CAZ, CTX, MRP	+ (2)	bla_{TEM}, bla_{SHV}, *intI1*
305c	*Ochrobactrum anthropi* (01)	CAZ, CTX, MRP, IPM	+ (1)	bla_{TEM}, bla_{SHV}, bla_{NDM}, *intI1*
310c, 314b/m	*Stenotrophomonas maltophilia* (02)	CAZ, CTX, MRP, IPM	NV	bla_{TEM}, bla_{SHV}, bla_{NDM}, *intI1*
356c	*Acinetobacter baumannii* (01)	CAZ, CTX, MRP, IPM	+ (4)	bla_{TEM}, bla_{SHV}, bla_{NDM}, *intI1*

n, number; CAZ, ceftazidime; CTX, cefotaxime; MRP, meropenem; IPM, imipenem; NV, no visible plasmid band; +, plasmid band present.
**Escherichia coli* serotypes (UT, untypable serotype; R, rough serotype; O2, O2 serotype).

The class 1 integron-integrase gene (*intI1*) was detected in 22 (38·6%) isolates, and 16 of these isolates were found to harbour the genes encoding carbapenemases and/or ESBLs. All carbapenemase gene-positive isolates showed the *intI1* gene. Integrons may be located on mobile elements such as plasmids and transposons, and play an important role in the adaptation of bacteria (Labbate *et al.* 2009). They have reservoir of antibiotic-resistance genes. Class 1 integrons on plasmids facilitate conjugative-mediated transfer of genetic material through site-specific recombination reaction mediated by the integron integrase IntI1 (Deng *et al.* 2015). However, bacterial conjugation in the present study failed to prove the conjugative movement of bla_{NDM} gene after three repetitive mating experiments.

Sequencing results revealed a single genotype of TEM (bla_{TEM-1}), KPC (blaKPC-2) and NDM (bla_{NDM-1}), and thus, indicated the clonal spread of bacteria from common/different niches. However, bla_{SHV-11} and bla_{SHV-28} genotypes were recorded among bla_{SHV}-harbouring bacterial isolates. Nucleotide sequences of the bacterial 16S rDNA (rRNA), bla_{NDM}, bla_{KPC}, bla_{SHV}, bla_{TEM} and *intI1* genes have been submitted to the NCBI GenBank database with the following accession numbers: MH411113.1, MH411114.1, MH411115.1, MH411118.1, MH411119.1, MH411120.1, MH411121.1, MH411122.1, MH411116.1, MH411117.1 and KR676527.1.

Here we report for the first time the bla_{NDM-1} gene in *S. maltophilia*, *A. baumannii* and *O. anthropi* isolates, and the bla_{KPC-2} gene in an *E. coli* isolate from goat meat in India.

Despite the extensive studies on emerging carbapenemases in Gram-negative bacteria from human and environmental sources, the reports on emerging carbapenemases from food animal products are scanty. The occurrence of carbapenemase genes (bla_{NDM}/bla_{VIM}/bla_{IMP}/bla_{OXA-48}) in *Enterobacteriaceae*, from limited food animal products (cow milk/chicken meat/pig offal), has been reported from India, Egypt, Algeria, China, Lebanon, Pakistan and Hong Kong (Ghatak *et al.* 2013; Abdallah *et al.* 2015; Yaici *et al.* 2016; Diab *et al.* 2017; He *et al.* 2017; Ahmad *et al.* 2018; Sapugahawatte *et al.* 2020). Such reports on carbapenemases from food sources attract global attention because the carbapenems are critically important in human medicine.

Diversely distributed carbapenemases among bacteria in different countries might have emerged because of the imported cases where the carrier bacteria are endemic. Population mobility may be another reason behind International spread of carbapenem-resistant organisms. In most of the developing countries, close interactions between humans and animals may lead to the inter-transmissions of antimicrobial-resistant micro-organisms (Doyle 2015).

Meat is a rich source of animal protein, which helps the micro-organisms to survive and grow. Although meat from healthy animals usually does not possess micro-organisms, it is highly susceptible to microbial contamination (Komba *et al.* 2012). Meat is usually contaminated during slaughter and handling processes in unhygienic conditions (Garedew *et al.* 2015). Most of the slaughterhouses in India are small, which are operated manually. Meat from these slaughterhouses is consumed by domestic retail markets. Though the meat consumption in India is on increase, the quality management is compromised in small slaughterhouses (Central Pollution Control Board 2017).

There was no epidemiological investigation that could trace the primary source of carbapenemase- and ESBL-producing bacteria from goat meat in the present study. However, based on the scientific knowledge on epidemiology and global spread of carbapenemase- and ESBL producers, it could be assumed that these bacteria might have spread in or contaminated the meat through handling or processing of animal carcasses with poor hygiene practices.

To the best of our knowledge, it is the first study reporting $bla_{\text{NDM-1}}$ and $bla_{\text{KPC-2}}$ genes in bacteria from retail goat meat in India. Presence of genetic determinants of carbapenemases and ESBLs in bacterial isolates from goat meat develops serious concerns about public health and food safety as meat could contain a wide variety of micro-organisms (usually the contaminants from humans or environmental sources) that could be transmitted to humans or animals through the food chain. Therefore, surveillance and monitoring of carbapenemase- and ESBL producers in all components of the food chain are urgently needed, with a thorough investigation of all positive cases so that the therapeutic efficacy of the 'last resort antimicrobials' such as carbapenems could be restored.

Materials and methods

Brief background of the bacterial isolates and their sources

A total of 57 non-duplicate Gram-negative bacterial isolates were used for this study. These bacterial isolates were recovered from 61 retail goat meat samples collected from local shops in the meat markets at Durg (20), Rajnandgaon (08), Bilaspur (12), Raipur (10) and Dhamtari (11) districts of Chhattisgarh state in the year 2014. About 1 g of each meat sample was triturated to make tissue homogenate in sterilized phosphate-buffered saline and used to inoculate into 10 ml of sterilized nutrient broth and incubated at 37°C for 24 h. The incubated broth suspension was streaked directly onto HiCrome ESBL agar (HiCrome ESBL agar base supplemented with HiCrome ESBL selective supplement) (HiMedia Pvt. Ltd., Mumbai, India) for selective isolation of β-lactamase-producing bacteria. The bacterial isolates were presumptively identified by cultural and standard biochemical characters and kept in the form of pure culture slants at 4°C for further use. Serotyping of *E. coli* isolates was carried out at National Salmonella and Escherichia Centre, Central Research Institute, Kasauli, Himachal Pradesh.

Identification of bacteria based on 16S rRNA gene sequences

Bacterial isolates were specifically identified based on their 16S rRNA gene sequences. For this, a single colony of each bacterial isolate was used to inoculate into 5 ml of nutrient broth and incubated at 37°C for 18 h. Broth suspension in 1·5 ml tubes was centrifuged at 10 000 ***g*** for 3 min. The bacterial cells were washed twice with nuclease-free water by centrifuging at 10 000 ***g*** for 3 min. Bacterial genomic DNA was isolated using GeneJET Genomic DNA Purification Kit (Thermo Fisher Scientific, Vilnius, Lithuania) and used as template for PCR amplification of 16S rRNA gene for specific identification of bacterial isolates. The universal primers used for PCR amplification of bacterial 16S rRNA gene segment are mentioned in Table 3. The PCR amplified products were excised in agarose gels and purified using GeneJET PCR Purification Kit (Thermo Fisher Scientific). Purified PCR products were subjected to bidirectional Sanger's sequencing commercially.

Antimicrobial susceptibility testing

In vitro antimicrobial susceptibility of isolates was tested by the broth dilution method using ceftazidime, cefotaxime, meropenem and imipenem (Clinical and Laboratory Standards Institute 2015). The results were interpreted in MIC or the lowest concentration of antibiotics that prevent bacterial growth. For testing of *S. maltophilia* and *O. anthropi* using antimicrobials such as cefotaxime, meropenem and imipenem, without specific CLSI criteria, a relevant criterion of non-*Enterobacteriaceae* was applied. *E. coli* ATCC 25922 was used as a reference strain for quality control of the procedure.

Plasmid DNA purification and plasmid features

To isolate the bacterial plasmid DNA, a single colony of each isolate was inoculated into 10 ml of Luria–Bertani (LB) broth in 50 ml tubes and incubated at 37°C for 12–15 h in an incubator shaker with agitation at 180 revolutions per min (Singh 2015). The bacterial cells were prepared by centrifuging 2 ml of incubated broth suspension at 10 000 ***g*** for 3 min. Plasmid DNA was isolated using GeneJET Plasmid Miniprep Kit (Thermo Fisher Scientific) and analysed for their physical characters (number and molecular size). The plasmids were stored at 4°C and used for screening of carbapenemase, ESBL and *intI1* gene by PCR.

Table 3 Primers used to amplify the ESBL, carbapenemase and class 1 integron-integrase genes (Singh *et al.* 2019)

Target gene	Primers	Amplicon size (bp)	References
16S rRNA	27F-5′AGAGTTTGATCMTGGCTCAG 3′ 1492R-5′CGGTTACCTTGTTACGACTT 3′	~1466	Lane (1991)
bla_{TEM}	F-5′AAAATTCTTGAAGACG 3′ R-5′ TTACCAATGCTTAATCA 3′	1073	Sharma *et al.* (2010)
bla_{SHV}	F-5′TCAGCGAAAAACACCTTG 3′ R-5′TCCCGCAGATAAATCACC 3′	472	Yasmin (2012)
	F-5′CTTTACTCGCCTTTATCGGC 3′ R-5′TTACCGACCGGCATCTTTCC 3′	982	Jones *et al.* (2009)
bla_{KPC}	F-5′CATTCAAGGGCTTTCTTGCTGC 3′ R-5′ACGACGGCATAGTCATTTGC 3′	538	Dallenne *et al.* (2010)
	F-5′ATGTCACTGTATCGCCGTCT 3′ R-5′TTTTCAGAGCCTTACTGCCC 3′	892	Jones *et al.* (2009)
bla_{NDM}	F-5′GGTTTGGCGATCTGGTTTTC 3′ R-5′CGGAATGGCTCATCACGATC 3′	621	Poirel *et al.* (2011)
	F-5′CATTTGCGGGGTTTTTAATG 3′ R-5′GATTGGGGGTGACGTGGT 3′	881	Singh *et al.* (2019)
intI1	F-5′ATCATCGTCGTAGAGACGTCGG 3′ R-5′GTCAAGGTTCTGGACCAGTTGC 3′	892	Reyes *et al.* (2003)

Molecular detection of β-lactamase and *intI1* gene

The carbapenemase (bla_{NDM} and bla_{KPC}), ESBL (bla_{TEM} and bla_{SHV}) and class 1 integron-integrase (*intI1*) genes were detected by PCR amplification using specific primers (Sigma-Aldrich Chemicals Pvt. Ltd., Bengaluru, India) (Table 3). For PCR amplifications, 100 ng (5 μl) of DNA was added to 50 μl reaction mixtures containing $0{\cdot}2 \times 10^{-3}$ mol l^{-1} of dNTPs, $0{\cdot}2 \times 10^{-6}$ mol l^{-1} of each primer and 2·5 U of *Taq* DNA polymerase (Sigma-Aldrich) in 1 × PCR buffer with $1{\cdot}875 \times 10^{-3}$ mol l^{-1} of $MgCl_2$. The *Shigella* sp. (accession number KR872630.1), *K. pneumoniae* ATCC BAA-1705, *K. pneumoniae* ATCC 700603 and bla_{TEM}-positive *E. coli* (available at laboratory) strains were used as the positive controls for bla_{NDM}, bla_{KPC}, bla_{SHV} and bla_{TEM} genes, respectively. *Escherichia coli* ATCC 25922 was used as negative control for all PCR amplification reactions.

Cloning and sequencing of the genes

The bla_{NDM}, bla_{KPC}, bla_{SHV}, bla_{TEM} and *intI1* genes were further amplified using *Pfu* DNA polymerase. Specific PCR products were excised in agarose gels and purified using GeneJET PCR Purification Kit (Thermo Fisher Scientific). The purified PCR products were then ligated into a pJET 1·2/blunt cloning vector (50 ng). Ligated products were transformed into *E. coli* $DH_5\alpha$ competent cells using TransformAid Bacterial Transformation Kit (Thermo Fisher Scientific). The competent cell suspension was then plated onto pre-warmed LB agar plates containing 50 μg ml^{-1} of ampicillin. Plates were incubated overnight at 37°C in upside down position. Positive transformant colonies were selected by performing colony PCR. Plasmid DNA was isolated and used for bidirectional Sanger's sequencing commercially. The sequence data were analysed using online BioEdit software. The specific nucleotide sequences were submitted to the GenBank database of the National Center for Biotechnology Information (NCBI), and the accession numbers were received.

Bacterial conjugation

Conjugation assay was performed by the broth mating method followed by the selection of transconjugants on a solid plate as reported earlier (Zhang *et al.* 2013b) with minor modifications. The bla_{NDM}-harbouring *A. baumannii*, *S. maltophilia* and *O. anthropi* isolates were selected as the donor. Streptomycin-resistant *E. coli* HB101 (Promega Corporation, Madison, WI) was used as the recipient. Before starting the experiment, susceptibility of donors and recipient was tested against ampicillin and streptomycin. Donors were allowed to grow overnight in LB broth containing ampicillin (100 μg ml^{-1}) by shaking at 200 revolutions per min in an incubator shaker. Similarly, the recipient *E. coli* HB101 strain was grown with streptomycin at the concentration of 30 μg ml^{-1} in LB broth. Donor and recipient cells were harvested by centrifugation and washed and resuspended in LB broth at 1·0 McFarland's standard density. Then, 50 μl of each donor was mixed with 50 μl of the recipient (in 1·5 ml tubes containing 0·9 ml of LB broth with antibiotics (20 μg ml^{-1} of streptomycin and 30 μg ml^{-1} of ampicillin). Both donor and recipient were further incubated at 37°C for 20 h without shaking. The mixture was immediately serially diluted 10-fold and plated onto MacConkey agar containing 30 μg ml^{-1} of streptomycin and 100 μg ml^{-1} of ampicillin to select the transconjugants. Transconjugants were differentiated from non-*E. coli* donors based on lactose fermentation in MacConkey agar (a differential medium) and by biochemical characteristics. They were subjected to detection of specific β-lactamase genes by PCR.

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Conflict of Interest

No conflict of interest declared.

References

Abdallah, H.M., Reuland, E.A., Wintermans, B.B., Al Naiemi, N., Koek, A., Abdelwahab, A.M., Ammar, A.M.,Mohamed, A.A. *et al.* (2015) Extended-spectrum beta-lactamases and/ or carbapenemases-producing *Enterobacteriaceae* isolated from retail chicken meat in Zagazig, Egypt. *PLoS One* **10**, e0136052. https://doi.org/10.1371/journal.pone.0136052.

Ahmad, K., Khattak, F., Ali, A., Rahat, S., Noor, S., Mahsood, N. and Somayya, R. (2018) Carbapenemases and extended-spectrum β-lactamase–producing multidrug-resistant *Escherichia coli* isolated from retail chicken in Peshawar: first report from Pakistan. *J Food Prot* **81**, 1339–1345.

Allen, H.K., Donato, J., Wang, H.H., Cloud-Hansen, K.A., Davies, J. and Handelsman, J. (2010) Call of the wild: antibiotic resistance genes in natural environments. *Nat Rev Microbiol* **8**, 251–259.

Berg, G., Eberl, L. and Hartmann, A. (2005) The rhizosphere as a reservoir for opportunistic human pathogenic bacteria. *Environ Microbiol* **7**, 1673–1685.

Bonardi, S. and Pitino, R. (2019) Carbapenemase-producing bacteria in food-producing animals, wildlife and environment: a challenge for human health. *Ital J Food Saf* **8**, 7956. http://doi.org/10.4081/ijfs.2019.7956.

Central Pollution Control Board. (2017) Revised comprehensive industry document on slaughter houses. CPCB document. Delhi: Central Pollution Control Board. Ministry of Environment, Forest and Climate Change, Government of India.

Clinical and Laboratory Standards Institute. (2015). *Performance Standards for Antimicrobial Susceptibility Testing; Twenty-Fifth Informational Supplement*. CLSI document M100–S25. Wayne, PA: Clinical and Laboratory Standards Institute.

Dallenne, C., Da Costa, A., Decre, D., Favier, C. and Arlet, G. (2010) Development of a set of multiplex PCR assays for the detection of genes encoding important β-lactamases in *Enterobacteriaceae*. *J Antimicrob Chemother* **65**, 490–495.

Deng, Y., Bao, X., Ji, L., Chen, L., Liu, J., Miao, J., Chen, D. and Bian, H. (2015) Resistance integrons: class 1, 2 and 3 integrons. *Ann Clin Microbiol Antimicrob* **14**, 45. http://doi.org/10.1186/s12941-015-0100-6.

Diab, M., Hamze, M., Bonnet, R., Saras, E., Madec, J.Y. and Haenni, M. (2017) OXA-48 and CTX-M-15 extended-spectrum beta-lactamases in raw milk in Lebanon: epidemic spread of dominant *Klebsiella pneumoniae* clones. *J Med Microbiol* **66**, 1688–1691.

Diyantoro and Wardhana, D.K. (2019) Risk factors for bacterial contamination of bovine meat during slaughter in ten Indonesian abattoirs. *Vet Med Int* **2019**, 1–6. https://doi.org/10.1155/2019/2707064

Doyle, M.E. (2015) Multidrug-resistant pathogens in the food supply. *Foodborne Pathog Dis* **12**, 261–279.

Garedew, L., Hagos, Z., Addis, Z., Tesfaye, R. and Zegeye, B. (2015) Prevalence and antimicrobial susceptibility patterns of Salmonella isolates in association with hygienic status from butcher shops in Gondar town, Ethiopia. *Antimicrob Resist Infect Control* **4**, 21. https://doi.org/10.1186/s13756-015-0062-7.

Ghatak, S., Singha, A., Sen, A., Guha, C., Ahuja, A., Bhattacherjee, U., Das, S., Pradhan, N.R. *et al.* (2013) Detection of New Delhi metallo-beta-Lactamase and extended-spectrum beta-lactamase genes in *Escherichia coli* isolated from mastitic milk samples. *Transbound Emerg Dis* **60**, 385–389.

He, T., Wang, Y., Sun, L., Pang, M., Zhang, L. and Wang, R. (2017) Occurrence and characterization of $bla_{\mathrm{NDM-5}}$-positive *Klebsiella pneumoniae* isolates from dairy cows in Jiangsu, China. *J Antimicrob Chemother* **72**, 90–94.

Higgins, C.S., Avison, M.B., Jamieson, L., Simm, A.M., Bennett, P.M. and Walsh, T.R. (2001b) Characterization, cloning and sequence analysis of the inducible *Ochrobactrum anthropi* AmpC β-lactamase. *J Antimicrob Chemother* **47**, 745–754.

Higgins, C.S., Murtough, S.M., Williamson, E., Hiom, S.J., Payne, D.J., Russell, A.D. *et al.* (2001a) Resistance to antibiotics and biocides among non-fermenting Gram-negative bacteria. *Clin Microbiol Infect* **7**, 308–315.

Jones, C.H., Tuckman, M., Keeney, D., Ruzin, A. and Bradford, P.A. (2009) Characterization and sequence analysis of extended-spectrum-β-lactamase-encoding genes from *Escherichia coli, Klebsiella pneumoniae,* and *Proteus mirabilis* isolates collected during tigecycline phase 3 clinical trials. *Antimicrob Agents Chemother* **53**, 465–475.

Jones, L.S., Toleman, M.A., Weeks, J.L., Howe, R.A., Walsh, T.R. and Kumarasamy, K.K. (2014) Plasmid carriage of *bla*$_{NDM-1}$ in clinical *Acinetobacter baumannii* isolates from India. *Antimicrob Agents Chemother* **58**, 4211–4213.

Komba, E.V.G., Komba, E.V., Mkupasi, E.M., Mbyuzi, A.O., Mshamu, S., Luwumba, D., Basagwe, Z. and Mzula, A. (2012) Sanitary practices and occurrences of zoonotic conditions in cattle at slaughter in Morogoro Municipality, Tanzania: implication for Public Health. *Tanzan J Health Res* **14**, 131–138.

Kumarasamy, K.K., Toleman, M.A., Walsh, T.R., Bagaria, J., Butt, F.A., Balakrishnan, R., Chaudhary, U., Doumith, M. *et al.* (2010) Emergence of a new antibiotic resistance mechanism in India, Pakistan, and the UK: a molecular, biological, and epidemiological study. *Lancet Infect Dis* **10**, 597–602.

Labbate, M., Case, R.J. and Stokes, H.W. (2009) The integron/gene cassette system: an active player in bacterial adaptation. *Methods Mol Biol* **532**, 103–125.

Lane, D.J. (1991) 16S/23S rRNA sequencing. In *Nucleic acid Techniques in Bacterial Systematic*, ed. Stackenbrandt, E. and Goodfellow, M. pp. 115–175. Chichester: John Wiley and Sons.

Lupo, A., Coyne, S. and Berendonk, T.U. (2012) Origin and evolution of antibiotic resistance: the common mechanisms of emergence and spread in water bodies. *Front Microbiol* **3**, 18. https://doi.org/10.3389/fmicb.2012.00018

Madec, J.Y., Haenni, M., Nordmann, P. and Poirel, L. (2017) Extended-spectrum β-lactamase/AmpC- and carbapenemase producing *Enterobacteriaceae* in animals: a threat for humans? *Clin Microbiol Infect* **23**, 826–833.

Martínez, J.L. (2008) Antibiotics and antibiotic resistance genes in natural environments. *Science* **321**, 365–367.

Morrill, H.J., Pogue, J.M., Kaye, K.S. and LaPlante, K.L. (2015) Treatment options for carbapenem-resistant *Enterobacteriaceae* infections. *Open Forum Infect Dis* **2**, ofv050. https://doi.org/10.1093/ofid/ofv050

Poirel, L., Revathi, G., Bernabeu, S. and Nordmann, P. (2011) Detection of NDM-1 producing *Klebsiella pneumoniae* in Kenya. *Antimicrob Agents Chemother* **55**, 934–936.

Pournaras, S., Koumaki, V., Spanakis, N., Gennjmata, V. and Tsakris, A. (2016) Current perspectives on tigecycline resistance in Enterobacteriaceae: susceptibility testing issues and mechanisms of resistance. *Int J Antimicrob Agents* **48**, 11–18.

Pruthvishree, B.S., Vinodh Kumar, O.R., Sinha, D.K., Malik, Y.P., Dubal, Z.B., Desingu, P.A., Shivakumar, M., Krishnaswamy, N. *et al.* (2017) Spatial molecular epidemiology of carbapenem-resistant and New Delhi metallo beta-lactamase (*bla*$_{NDM}$)-producing *Escherichia coli* in the piglets of organized farms in India. *J Appl Microbiol* **122**, 1537–1546.

Rameshkumar, M.R., Vignesh, R., Swathirajan, C.R., Balakrishnan, P. and Arunagirinathan, N. (2015) Chromogenic agar medium for rapid detection of extended-spectrum β-lactamases and *Klebsiella pneumoniae* carbapenemases producing bacteria from human immunodeficiency virus patients. *J Res Med Sci* **20**, 1219–1220.

Reyes, A., Bello, H., Domínguez, M., Mella, S., Zemelman, R., and González, G. (2003) Prevalence and types of class 1 integrons in aminoglycoside-resistant *Enterobacteriaceae* from several Chilean hospitals. *J Antimicrob Chemother* **51**, 317–321.

Sanchez, M.B. (2015) Antibiotic resistance in the opportunistic pathogen *Stenotrophomonas maltophilia*. *Front Microbiol* **6**, 658. https://doi.org/10.3389/fmicb.2015.00658

Sanchez, M.B., Hernandez, A. and Martinez, J.L. (2009) *Stenotrophomonas maltophilia* drug resistance. *Future Microbiol* **4**, 655–660.

Sapugahawatte, D.N., Li, C., Zhu, C., Dharmaratne, P., Wong, K.T., Lo, N. and Ip, M. (2020) Prevalence and characteristics of extended-spectrum-β-lactamase-producing and carbapenemase-producing Enterobacteriaceae from freshwater fish and pork in wet markets of Hong Kong. *mSphere* **5**, e00107-20. https://doi.org/10.1128/mSphere.00107-20.

Scotta, C., Juan, C., Cabot, G., Oliver, A., Lalucat, J., Bennasar, A. and Sebastián, A. (2011) Environmental microbiota represents a natural reservoir for dissemination of clinically relevant metallo-β-lactamases. *Antimicrob Agents Chemother* **55**, 5376–5379.

Sharma, J., Sharma, M. and Ray, P. (2010) Detection of TEM & SHV genes in *Escherichia coli* & *Klebsiella pneumoniae* isolates in a tertiary care hospital from India. *Indian J Med Res* **132**, 332–336.

Singh, F. (2015) *Studies on extended-spectrum beta-lactamases and carbapenemases coding genes that confer multi-drug resistance to enteropathogenic bacteria.* PhD Thesis, Chhattisgarh Kamdhenu Vishwavidyalaya, Durg.

Singh, F., Hirpurkar, S.D., Shakya, S., Rawat, N., Devangan, P., Khan, F.F. and Bhandekar, S.K. (2017) Presence of enterobacteria producing extended-spectrum β-lactamases and/or carbapenemases in animals, humans and environment in India. *Thai J Vet Med* **47**, 35–43.

Singh, F., Hirpurkar, S.D., Rawat, N., Shakya, S., Kumar, R., Kumar, S., Meena, R.K., Rajput, P.K. *et al.* (2019) Carbapenemase and ESBL genes with class 1 integron among fermenting and nonfermenting bacteria isolated from water sources from India. *Lett Appl Microbiol* **71**, 70–77.

Tanganyika, J., Mfitilodze, W.M., Mtimuni, J.P. and Phoya, R.R.K.D. (2017) Microbial quality of goat carcasses in Lilongwe, Malawi. *Chem Biol Technol Agric* **4**, 27. http://doi.org/10.1186/s40538-017-0109-5.

Tepeli, S.O. and Demirel Zorba, N.N. (2017) Frequency of extended-spectrum β-lactamase (ESBL)– and AmpC β-lactamase–producing *Enterobacteriaceae* in a cheese production process. *J Dairy Sci* **101**, 2906–2914.

van Duin, D. and Doi, Y. (2017) The global epidemiology of carbapenemase-producing *Enterobacteriaceae*. *Virulence* **8**, 460–469.

Woodford, N., Wareham, D.W., Guerra, B. and Teale, C. (2014) Carbapenemase-producing *Enterobacteriaceae* and non-*Enterobacteriaceae* from animals and the environment: an emerging public health risk of our own making? *J Antimicrob Chemother* **69**, 287–291.

Yaici, L., Haenni, M., Saras, E., Boudehouche, W., Touati, A. and Madec, J.Y. (2016) $bla_{\mathrm{NDM\text{-}5}}$ carrying IncX3 plasmid in *Escherichia coli* ST1284 isolated from raw milk collected in a dairy farm in Algeria. *J Antimicrob Chemother* **71**, 2671–2672.

Yasmin, T. (2012) *Prevalence of ESBL among Escherichia coli and Klebsiella spp. in a tertiary care hospital and molecular detection of important ESBL producing genes by multiplex PCR*. MD Thesis, Dhaka University.

Zhang, C., Qiu, S., Wang, Y., Qi, L., Hao, R., Liu, X., Shi, Y., Hu, X. *et al.* (2013a) Higher isolation of NDM-1 producing *Acinetobacter baumannii* from the sewage of the hospitals in Beijing. *PLoS One* **8**, e64857. https://doi.org/10.1371/journal.pone.0064857.

Zhang, P.Y., Xu, P.P., Xia, Z.J., Wang, J., Xiong, J. and Li, Y.Z. (2013b) Combined treatment with the kanamycin and streptomycin promotes the conjugation of *Escherichia coli*. *FEMS Microbiol Lett* **348**, 149–156.